# ⚙️ 4-Stroke Theory & Engine Parts: Reading

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

## How a Four-Stroke Engine Works

A small gas engine turns the energy in gasoline into motion. Most lawn mowers and many other small engines use a four-stroke cycle. A stroke is one movement of the piston, up or down, inside the cylinder.

On the intake stroke, the piston moves down and the intake valve opens, pulling in a mixture of air and fuel. On the compression stroke, both valves close and the piston moves up, squeezing the mixture. On the power stroke, the spark plug fires, the mixture burns and expands, and the piston is forced down. On the exhaust stroke, the exhaust valve opens and the piston pushes the burned gases out.

The piston is connected to the crankshaft by the connecting rod. The crankshaft changes up-and-down motion into rotation. It turns twice for every four strokes, and the camshaft, turning at half speed, opens each valve at the right moment. A heavy flywheel keeps the engine turning between power strokes.

Two-stroke engines, used in chainsaws and trimmers, complete the cycle in one crankshaft turn. They are lighter and make power every turn, but their oil is mixed into the fuel and burned, so they create more pollution.

## Check your reading

1. List the four strokes in order.

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_____

2. What does the crankshaft do?

_____

_____

3. Give one difference between 2-stroke and 4-stroke engines.

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